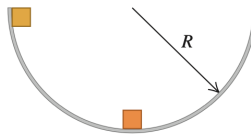


Problem set 5 (due April 20)

- (20pts) A 45kg person stands up in a 60kg canoe that is 5m long. The person walks from a point 1m from one end to a point 1m from the other end. If you ignore resistance to motion of the canoe in the water, how far does the canoe move during this process?
- (20pts) A 12kg ball is launched at an angle of 55 degrees above the horizontal with an initial speed of 150m/s . At its highest point, the ball explodes into two fragments, one three times heavier than the other. The two fragments reach the ground at the same time. Assume there is no air resistance. If the heavier fragment lands back at the point from which the ball was launched, find:
 - (10pts) The location where the lighter fragment lands
 - (10pts) How much energy is released in the explosion
- (35pts) Consider two balls of masses m and $M = 2m$. At time $t = 0$, you release the ball of mass M from a distance $h = 20\text{m}$ above the Earth's surface. One second later, you release the second ball of mass m from the same distance h above the ground. Ignore air resistance and assume all collisions are elastic.
 - (5pts) Find the time t that it takes the ball of mass M to reach the ground.
 - (10pts) Find the time and the distance above the ground where the first collision between the balls takes place.
 - (10pts) Compute the velocities (speed and direction) of the two balls right after the first collision.
 - (10pts) Find the maximum distance above the ground reached by the ball of mass m after the collision.
- (25pts) Two identical masses are released from rest in a smooth and frictionless hemispherical bowl of radius R as shown in the figure below.



- (15pts) If the masses stick together when they collide, how high above the bottom of the bowl will they go after colliding?
- (10pts) Compare the initial and final gravitational potential energies and explain the result.